BMC Cardiovascular Disorders

RESEARCH

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# Machine learning-based prediction of atherosclerotic cardiovascular disease risk in adults with diabetes or prediabetes

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## Abstract

**Background** Diabetes and prediabetes significantly increase the risk of atherosclerotic cardiovascular disease (ASCVD), posing a major global health challenge. Although traditional ASCVD risk factors have been extensively studied, there is limited research on applying machine learning.

**Methods** This study used data from the NHANES survey spanning 2007 to 2018, including 4,211 participants diagnosed with diabetes or prediabetes. Key variables were identified through univariate and multivariate logistic regression analyses. The dataset was randomly split into training and validation sets at a 7:3 ratio. Nine machine learning models (including CART, SVM, and GBM) were developed and evaluated using AUC, Brier scores, calibration curves, and decision curve analysis. Additionally, an online risk prediction platform was created to provide real-time ASCVD risk assessments, helping clinicians with early screening and intervention.

**Results** Eight variables significantly associated with ASCVD risk were identified through univariate and multivariate logistic regression analyses, including age, waist circumference, poverty–income ratio, blood urea nitrogen, total cholesterol, systolic blood pressure, hypertension, and smoking status. Based on these predictors, the SVM model achieved AUC values of 0.831 in the training set and 0.859 in the validation set, demonstrating excellent discriminative ability. Calibration curves indicated good agreement between predicted and observed risks across different risk levels, while Brier scores further supported the overall predictive accuracy of the model. Decision curve analysis showed that the model provided substantial net clinical benefit. Collectively, these evaluation metrics highlight the strong predictive performance and practical utility of the SVM model.

**Conclusion** The SVM model effectively predicts ASCVD risk in individuals with diabetes or prediabetes, emphasizing the importance of managing modifiable risk factors such as dyslipidemia, hypertension, smoking, and abdominal obesity. This model offers an effective tool for individualized risk assessment and early prevention, and the online platform further supports clinical application by enabling early identification and intervention for high-risk populations.

**Keywords** Diabetes, Prediabetes, Atherosclerotic cardiovascular disease, Machine learning, Predictive model

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## Introduction

Diabetes mellitus is one of the most prevalent chronic metabolic diseases worldwide [1]. It is characterized by persistent hyperglycemia, primarily caused by insufficient insulin secretion, and is closely associated with various complications, including cardiovascular diseases and renal failure [2–5]. Epidemiological data indicate that approximately 13% of adults in the United States have diabetes [6]. According to the International Diabetes Federation, the global prevalence of diabetes has increased significantly in recent years and is projected to reach 700 million by 2045 [7, 8].

Prediabetes is a metabolic condition between normoglycemia and diabetes, mainly comprising impaired fasting glucose and impaired glucose tolerance [5]. As defined by the American Diabetes Association (ADA), diagnostic criteria for prediabetes include: fasting plasma glucose between 5.6 and 6.9 mmol/L, 2-hour post-load plasma glucose between 7.8 and 11.0 mmol/L, or glycated hemoglobin (HbA1c) levels between 39 and 47 mmol/mol [9, 10]. Studies have shown that prediabetes is associated with a significantly increased risk of both microvascular and macrovascular complications [10, 11].

Atherosclerotic Cardiovascular Disease (ASCVD) refers to a group of common conditions caused by atherosclerosis, including coronary heart disease and stroke. It is characterized by arterial wall thickening and reduced elasticity, leading to cardiovascular dysfunction [12]. Epidemiological studies indicate that, driven by factors such as population aging, the global prevalence of ASCVD continues to rise, resulting in substantial mortality and economic burden [13].

Dyslipidemia is a well-established contributor to atherosclerosis, with abnormal levels of total cholesterol (TCHOL), high-density lipoprotein cholesterol (HDL), and triglycerides (TG) playing important roles in vascular pathology. Elevated TCHOL and TG, along with reduced HDL levels, are associated with lipid accumulation in arterial walls, promoting endothelial dysfunction, foam cell formation, and plaque development [14–16]. These lipid disturbances, combined with chronic inflammation and other metabolic risk factors, accelerate the progression of ASCVD [12, 17].

Clinically, coronary heart disease typically presents with exertional chest pain that may intensify under physical or emotional stress, potentially leading to myocardial infarction [18]. Stroke, depending on its ischemic or hemorrhagic nature, manifests as neurological impairments such as hemiplegia or aphasia [19, 20].

Machine learning is a powerful analytical tool that has been widely applied in the construction of risk prediction models in the medical field [21–25]. It demonstrates unique advantages in handling complex medical data by identifying potential patterns and associations among a large number of variables that are often overlooked by traditional statistical methods [26]. This capability often surpasses the predictive accuracy of conventional approaches, providing clinicians with more precise risk assessments.

Given the strong link between diabetes or prediabetes and ASCVD, early identification of high-risk individuals is essential. Using NHANES data (2007–2018), we first performed univariate logistic regression to screen candidate variables associated with ASCVD, followed by multivariable logistic regression to identify independent predictors. These predictors were then incorporated into a support vector machine (SVM) model to develop and validate a machine learning–based risk prediction tool. We further deployed an online risk calculator to facilitate preliminary ASCVD risk estimation and individualized stratification in patients with diabetes or prediabetes.

## Methods

### Data source and participants

The data for this study were obtained from the National Health and Nutrition Examination Survey (NHANES) conducted between 2007 and 2018. NHANES is a nationally representative cross-sectional survey that employs a multistage, stratified probability sampling design to systematically assess the health and nutritional status of the U.S. population. All participants provided written informed consent, and the study was approved by the Institutional Review Board of the National Center for Health Statistics.

Participants were considered eligible if they were aged 20 years or older and had completed demographic interviews, physical examinations, and questionnaires. From this cohort, individuals with a diagnosis of diabetes or prediabetes were further identified as the target population. During the screening process, we excluded participants younger than 20 years ($n = 2{,}887$), those without a diagnosis of diabetes or prediabetes ($n = 38{,}340$), and those with missing core covariate information among the diabetes/prediabetes group ($n = 4{,}715$). After applying these criteria, a total of 4,211 participants were included in the final analysis. The detailed selection process is illustrated in Fig. 1.

### Data extraction and variable measurement

Questionnaire data in NHANES were collected using standardized procedures by trained interviewers through the Computer-Assisted Personal Interviewing system. Key variables such as ASCVD, diabetes, and hypertension were primarily self-reported and, when applicable, cross-validated with laboratory or physical measurements (e.g., HbA1c, blood pressure) to enhance data reliability.

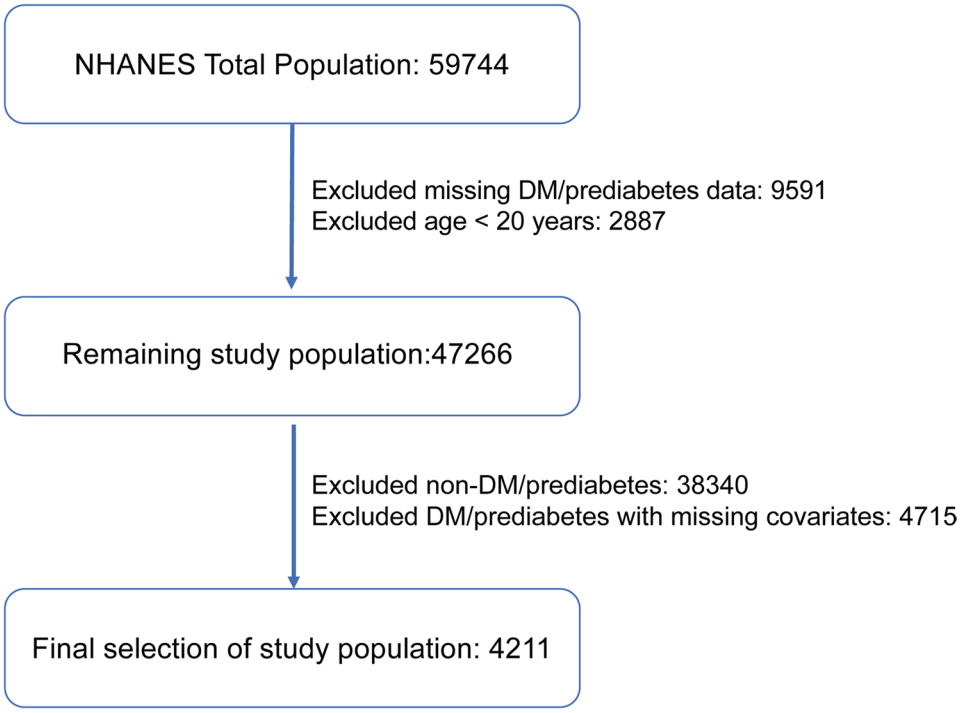



**Fig. 1** Flow chart for the study population

***Definition of prediabetes and diabetes***

Prediabetes was identified by asking participants whether they had ever been informed by a healthcare professional that they had prediabetes, impaired fasting glucose, impaired glucose tolerance, borderline diabetes, or elevated blood sugar not reaching diabetic thresholds. An affirmative response classified the participant as having prediabetes. Diabetes was identified by the question, "Have you ever been told by a doctor or health professional that you have diabetes?" In addition to self-reported diagnoses, HbA1c levels were used for classification. According to the ADA criteria, an HbA1c level of 5.7%–6.4% (39–47 mmol/mol) defined prediabetes, and ≥ 6.5% (48 mmol/mol) defined diabetes.

***Definition of ASCVD***

ASCVD was defined as a self-reported physician diagnosis of coronary heart disease, angina, myocardial infarction, or stroke, with myocardial infarction and stroke considered hard endpoints. Participants who answered "yes" to the question "Has a doctor or other health professional ever told you that you had coronary heart disease, angina, heart attack, or stroke?" were classified as having ASCVD.

***Measurement and definition of other variables***

Candidate predictors were selected based on prior evidence, clinical relevance, and professional guidelines, following a strategy similar to that used by Khan et al. (2024) in developing the AHA PREVENT equations [27]. We included traditional ASCVD risk factors (e.g., age, sex, race/ethnicity, smoking, blood pressure, lipids) as well as additional variables reported in the literature to be associated with ASCVD risk in individuals with diabetes, such as socioeconomic status and lifestyle behaviors. In total, 26 predictors were retained. Demographic variables included age, sex, race/ethnicity, marital status, and education level. The poverty-income ratio (PIR) was calculated as the ratio of annual household or personal income to the poverty threshold for the participant's survey year and region, and was treated as a continuous variable.

Smoking status was self-reported and categorized as never, former, or current smoker. Drinking status was classified according to NHANES criteria as never, former, light, moderate, or heavy. Lifestyle variables included average weekday sleep duration, daily sedentary time, and weekly physical activity(PA), all assessed via questionnaire. Hypertension status was defined as a binary variable based on self-reported physician diagnosis.

Anthropometric data were obtained by trained staff at NHANES Mobile Examination Centers, including waist circumference (WC), systolic blood pressure (SBP), diastolic blood pressure (DBP), and body mass index (BMI). BMI was calculated as weight in kilograms divided by the square of height in meters ($kg/m^2$) and classified according to Centers for Disease Control and Prevention (CDC) guidelines as normal (< 25 $kg/m^2$), overweight

(25–29.9 kg/m²), or obese (≥ 30 kg/m²). Abdominal obesity was defined as WC ≥ 102 cm in men and ≥ 88 cm in women, based on WHO guidelines.

We included continuous variables such as creatinine (CR), uric acid (UA), albumin (ALB), blood urea nitrogen (BUN), TG, alanine aminotransferase (ALT), aspartate aminotransferase (AST), TCHOL, HDL, and HbA1c, as these indicators have been widely recognized in prior studies as relevant markers of cardiovascular and metabolic risk.

### Data preprocessing

This study was based on publicly available data from six consecutive 2-year cycles of NHANES (2007–2018). According to NHANES analytical guidelines, combined sample weights were calculated by dividing the original 2-year weights by 6 to ensure nationally representative estimates. Continuous variables were standardized using Z-score transformation, and missing values were imputed using the median for continuous variables and the mode for categorical variables. Data preprocessing, including data wrangling and imputation, was performed using the "dplyr" and "tidyr" packages in R.

The dataset was randomly split into training and validation(test) sets at a ratio of 7:3, with stratified sampling based on ASCVD status to ensure balanced outcome distribution. This was implemented using the "caret" package.

Weighted univariate logistic regression analyses were conducted to identify ASCVD-related predictors using the "survey" package, and variables with $P < 0.05$ were further assessed in multivariate analyses. Significant predictors were subsequently included as features for machine learning modeling.

### Machine learning model construction

Supervised machine learning models were developed using the "caret" package in R. A total of nine algorithms were applied, including Logistic Regression (LR), Classification and Regression Tree (CART), Support Vector Machine (SVM), Gradient Boosting Machine (GBM), k-Nearest Neighbors (KNN), Adaptive Boosting (AdaBoost), Linear Discriminant Analysis (LDA), Neural Network (NNET), and Extreme Gradient Boosting (XGBoost). All models were trained on the training dataset using 5-fold cross-validation to reduce overfitting, with optimization based on the area under the receiver operating characteristic curve (AUC).

For the CART model, the complexity parameter was set to 0.001. The XGBoost model employed automated hyperparameter tuning with tuneLength set to 5, while default parameters were adopted for the other models. All procedures for model training, tuning, and validation were performed within the "caret" framework.

Model performance was comprehensively assessed from four aspects: discriminative ability was evaluated using AUC with the "pROC" package; calibration was assessed using calibration curves constructed via the "rms" package; clinical utility was assessed by decision curve analysis (DCA) utilizing the "rmda" package; and probabilistic accuracy was evaluated by Brier scores.

Variable importance was analyzed using the "varImp" function in "caret", which computes model-specific estimates based on each variable's contribution to classification performance. For models such as SVM, importance was derived from internal model coefficients. The relative importance of predictors was visualized to facilitate clinical interpretation.

### Development and deployment of the online ASCVD risk prediction tool

An online prediction tool was developed and deployed based on the R Shiny framework. The tool integrates the best-performing machine learning model to generate real-time risk predictions for ASCVD based on user-inputted individual variables. To enhance model interpretability, the tool provides visualizations of key features, aiming to offer convenient decision support for clinicians and public health personnel.

### Statistical analysis

For baseline characteristics, continuous variables were presented as weighted means ± standard deviations, while categorical variables were presented as weighted frequencies and percentages. Group comparisons were performed using t-tests for continuous variables and chi-square or Fisher's exact tests for categorical variables. Results were reported as odds ratios (ORs) with corresponding 95% confidence intervals (CIs). All statistical tests were two-sided, with a significance threshold set at $P < 0.05$. All statistical analyses were conducted using R software (version 4.4.3). The evaluation of model performance, including discrimination, calibration, clinical utility, and accuracy, has been detailed in the previous sections. This study adhered to the TRIPOD guidelines and followed the TRIPOD-AI recommendations where applicable for machine learning methods.

## Results

### Baseline information

Compared to the non-ASCVD group, the ASCVD group exhibited a less favorable cardiometabolic profile, including higher age, SBP, WC, and BMI; significantly higher CR, BUN, and HbA1c levels; slightly higher TG; and lower HDL cholesterol and ALB levels. Additionally, the ASCVD group had a lower PIR, longer sedentary time, and shorter total PA time (Table 1).

**Table 1** Baseline information on the total population

| Variable | Total | No-ASCVD | ASCVD | *P*-value |
|---|---|---|---|---|
| **n** | 4211 | 3552 | 659 | |
| **SBP (mmHg)** | 128.2 (18.7) | 127.5 (18.4) | 131.8 (19.8) | < 0.001 |
| **DBP (mmHg)** | 69.9 (11.1) | 69.8 (10.9) | 70.0 (12.1) | 0.776 |
| **Sleep (hours)** | 7.0 (1.5) | 7.0 (1.5) | 7.1 (1.7) | 0.367 |
| **PIR** | 2.5 (1.6) | 2.6 (1.6) | 2.3 (1.6) | 0.001 |
| **WC (cm)** | 106.3 (16.1) | 105.7 (16.1) | 109.8 (15.6) | < 0.001 |
| **Age (years)** | 56.5 (14.9) | 54.8 (14.9) | 65.8 (10.9) | < 0.001 |
| **BMI (kg/m2)** | 31.3 (7.0) | 31.2 (7.0) | 31.8 (7.1) | 0.029 |
| **Sedentary (minutes)** | 337.4 (194.7) | 332.9 (194.9) | 361.9 (191.6) | < 0.001 |
| **CR (mg/dl)** | 0.9 (0.5) | 0.9 (0.5) | 1.1 (0.7) | < 0.001 |
| **PA_total_time (minutes)** | 547.3 (804.6) | 568.8 (829.9) | 431.4 (639.7) | < 0.001 |
| **UA (mg/dl)** | 5.8 (1.5) | 5.7 (1.5) | 6.1 (1.5) | 0.091 |
| **ALB (mg/dl)** | 4.2 (0.3) | 4.2 (0.3) | 4.1 (0.3) | < 0.001 |
| **BUN (mg/dl)** | 14.7 (6.5) | 14.2 (6.1) | 17.3 (7.9) | < 0.001 |
| **TG (mg/dl)** | 176.5 (158.3) | 176.2 (162.2) | 178.3 (135.3) | 0.047 |
| **ALT (mg/dl)** | 27.7 (21.5) | 28.0 (18.3) | 26.1 (33.7) | 0.059 |
| **AST (mg/dl)** | 26.7 (15.6) | 26.7 (15.0) | 26.8 (18.5) | 0.864 |
| **TCHOL (mg/dl)** | 181.8 (40.6) | 181.4 (40.3) | 183.8 (41.9) | 0.168 |
| **HDL (mg/dl)** | 49.5 (14.9) | 49.9 (14.9) | 47.4 (14.7) | < 0.001 |
| **HbA1c** | 6.6 (1.6) | 6.6 (1.6) | 6.8 (1.5) | 0.008 |
| **Sex** | | | | < 0.001 |
| Female | 1790 (42.5) | 1560 (43.9) | 230 (34.9) | |
| Male | 2421 (57.5) | 1992 (56.1) | 429 (65.1) | |
| **Race** | | | | 0.052 |
| Non-Hispanic White | 1684 (40.0) | 1353 (38.1) | 331 (50.2) | |
| Non-Hispanic Black | 877 (20.8) | 751 (21.1) | 126 (19.1) | |
| Mexican American | 718 (17.1) | 638 (18.0) | 80 (12.1) | |
| Other Hispanic | 448 (10.6) | 387 (10.9) | 61 (9.3) | |
| Other Race - Including Multi-Racial | 484 (11.5) | 423 (11.9) | 61 (9.3) | |
| **Marital** | | | | < 0.001 |
| Never married | 485 (11.5) | 454 (12.8) | 31 (4.7) | |
| Married/Living with Partner | 2661 (63.2) | 2263 (63.7) | 398 (60.4) | |
| Widowed/Divorced/Separated | 1065 (25.3) | 835 (23.5) | 230 (34.9) | |
| **Education** | | | | 0.013 |
| Less Than 9th Grade | 491 (11.7) | 395 (11.1) | 96 (14.6) | |
| 9-11th Grade (Includes 12thgrade with no diploma) | 595 (14.1) | 505 (14.2) | 90 (13.7) | |
| High school graduate/GED or equivalent | 999 (23.7) | 827 (23.3) | 172 (26.1) | |
| Some college or AA degree | 1245 (29.6) | 1059 (29.8) | 186 (28.2) | |
| College graduate or above | 881 (20.9) | 766 (21.6) | 115 (17.5) | |
| **Alcohol.user** | | | | < 0.001 |
| never | 631 (15.0) | 544 (15.3) | 87 (13.2) | |
| former | 808 (19.2) | 641 (18.0) | 167 (25.3) | |
| mild | 1519 (36.1) | 1246 (35.1) | 273 (41.4) | |
| moderate | 530 (12.6) | 463 (13.0) | 67 (10.2) | |
| heavy | 723 (17.2) | 658 (18.5) | 65 (9.9) | |
| **Smoke status** | | | | < 0.001 |
| Never | 2168 (51.5) | 1903 (53.6) | 265 (40.2) | |
| Former | 1321 (31.4) | 1039 (29.3) | 282 (42.8) | |
| Now | 722 (17.1) | 610 (17.2) | 112 (17.0) | |
| **Hypertension status** | | | | < 0.001 |

**Table 1** (continued)

| Variable | Total | No-ASCVD | ASCVD | *P*-value |
|---|---|---|---|---|
| No | 1624 (38.6) | 1506 (42.4) | 118 (17.9) | |
| Yes | 2587 (61.4) | 2046 (57.6) | 541 (82.1) | |

Data are presented as mean (standard deviation) for continuous variables and n (%) for categorical variables

*Abbreviations*: *Ref* Reference, *SBP* Systolic Blood Pressure, *DBP* Diastolic Blood Pressure, *PIR* Poverty Income Ratio, *WC* Waist Circumference, *BMI* Body Mass Index, *CR* Creatinine, *PA* Physical Activity, *UA* Uric Acid, *ALB* Albumin, *BUN* Blood Urea Nitrogen, *TG* Triglycerides, *ALT* Alanine Aminotransferase, *AST* Aspartate Aminotransferase, *TCHOL* Total Cholesterol, *HDL* High-Density Lipoprotein, *HbA1c* Glycated Hemoglobin

For categorical variables, the ASCVD group had a higher proportion of males, a significantly higher prevalence of hypertension, more former smokers and fewer never smokers, and higher proportions of light and former drinkers with fewer heavy drinkers. The ASCVD group also had more widowed/divorced/separated individuals and fewer never-married individuals, with overall lower educational attainment.

In the training set (Supplementary Table S1), the ASCVD group had significantly higher age, SBP, WC, BMI, CR, BUN, and HbA1c levels, with significantly less total physical activity. Categorical variables showed higher proportions of males, hypertension, former smokers, light drinkers, and widowed/divorced/separated individuals, with lower education levels.

In the validation set (Supplementary Table S2), ASCVD patients showed significant differences ($P<0.05$) from the non-ASCVD group in age, SBP, WC, CR, BUN, UA, and total physical activity. For categorical variables, ASCVD patients had higher proportions of males, higher hypertension prevalence, more former smokers, and higher proportions of light drinkers, with more widowed/divorced/separated individuals.

#### Univariate and multivariate logistic regression analyses

Supplementary Table S3 presents the results of univariate logistic regression analysis, in which a broad range of variables, including demographic, clinical, and laboratory factors, showed significant associations with ASCVD risk ($P< 0.05$). These included age, WC, systolic and DBP, PIR, CR, UA, ALB, BUN, TCHOL, HbA1c, physical activity time, sleep duration, marital status, smoking status, hypertension status, race, education level, and alcohol consumption.

Table2 shows the results of multivariate logistic regression, where age, WC, SBP, PIR, BUN, TCHOL, smoking status, and hypertension status remained independent predictors ($P< 0.05$). These significant predictors from the multivariate model were subsequently used as input features in machine learning models.

#### Development and validation of the model

This study constructed nine supervised learning models (LR, CART, SVM, GBM, KNN, AdaBoost, LDA, NNET, XGBoost) based on eight clinical variables (age, WC, SBP, TCHOL, PIR, hypertension status, smoking status, and BUN), and comprehensively evaluated them from four dimensions: discrimination (AUC), calibration, Brier score, and DCA.

In terms of discrimination (Figure 2), SVM's AUC was the highest in the training set, followed by KNN and GBM (Figure 2A); in the validation set, SVM's AUC was 0.859, while KNN's and XGBoost's were 0.855 and 0.852, respectively, with the three being close (Figure 2B). Threshold-dependent metrics show that SVM is more balanced between sensitivity and specificity (0.790/0.696), while KNN and XGBoost were 0.771/0.683 and 0.781/0.685, respectively (Table 3). In comparison, GBM's sensitivity was acceptable (0.763), but its specificity was relatively low (0.641), presenting a typical recall-specificity trade-off (a reduction in missed diagnoses accompanied by an increase in false alarms).

The calibration analysis (Figure 3) shows that the calibration curves of SVM and XGBoost have a high degree of overlap with the ideal reference line in the mid-to-high risk range; KNN's calibration curve is mostly below the ideal line, suggesting a tendency to overestimate risk.

Overall predictive error was measured by the Brier score (Table 4), an indicator that reflects the mean squared error between predicted probabilities and actual outcomes, where lower values indicate smaller overall prediction error. CART had the lowest Brier score (Training/Validation: 0.096/0.093), but its performance in discrimination and calibration was average, limiting its practical value. The Brier scores for XGBoost and SVM were 0.111/0.099 and 0.119/0.106, respectively, both at a good level.

DCA assesses the clinical utility of a model by its net benefit at different threshold probabilities (the higher the net benefit, the greater the clinical benefit). As shown in Figure 4, within a wide range of thresholds, the net benefit of SVM and XGBoost were both significantly higher than the 'treat-all' and 'treat-none' strategies; in the validation set, the net benefit curve of SVM was higher overall, with XGBoost following closely behind.

Based on the combined evidence (leading AUC in the validation set, a more balanced sensitivity and specificity, good calibration, and higher net benefit in the main threshold range), this study selected SVM as the primary model for subsequent analysis. To improve interpretability, we performed feature importance ranking for the SVM model (Figure 5), and the results showed that

**Table 2** Multivariate logistic regression analysis

| Character | *P* value | OR | 95% CI |
|---|---|---|---|
| **SBP (mmHg)** | 0.002 | 1.28 | (1.21,1.33) |
| **DBP (mmHg)** | 0.16 | 1.01 | (0.98,1.02) |
| **Sleep (hours)** | 0.34 | 1.04 | (0.96,1.13) |
| **PIR** | 0.004 | 0.85 | (0.81,0.96) |
| **WC (cm)** | 0.002 | 1.06 | (1.01,1.09) |
| **Age (years)** | < 0.001 | 1.27 | (1.19,1.33) |
| **CR (mg/dl)** | 0.13 | 1.56 | (1.43,1.67) |
| **PA_total_time (minutes)** | 0.38 | 1 | (0.98,1.01) |
| **UA (mg/dl)** | 0.52 | 1.01 | (0.91,1.13) |
| **ALB (mg/dl)** | 0.35 | 0.87 | (0.59,1.30) |
| **BUN (mg/dl)** | 0.04 | 1.12 | (1.08,1.17) |
| **TCHOL (mg/dl)** | 0.01 | 1.32 | (1.22,1.41) |
| **HbA1c** | 0.17 | 1.07 | (0.97,1.17) |
| **Marital** | | | |
| Never married | ref | ref | ref |
| Married/Living with Partner | 0.12 | 1.95 | (1.52,2.33) |
| Widowed/Divorced/Separated | 0.08 | 1.77 | (1.61,2.27) |
| **Education** | | | |
| Less Than 9th Grade | ref | ref | ref |
| 9-11th Grade (Includes 12th grade with no diploma) | 0.49 | 0.94 | (0.60,1.47) |
| High school graduate/GED or equivalent | 0.75 | 0.93 | (0.61,1.43) |
| Some college or AA degree | 0.13 | 0.95 | (0.62,1.47) |
| College graduate or above | 0.11 | 0.75 | (0.46,1.23) |
| **Alcohol.user** | | | |
| Never | ref | ref | ref |
| Former | 0.26 | 1.06 | (0.73,1.53) |
| Mild | 0.34 | 1.18 | (0.93,1.39) |
| Moderate | 0.71 | 1.61 | (1.41,1.82) |
| Heavy | 0.2 | 1.13 | (1.05,1.21) |
| **Smoke** | | | |
| Never | ref | ref | ref |
| Former | 0.01 | 1.24 | (1.09,1.4) |
| Now | 0.04 | 1.55 | (1.24,1.84) |
| **Hypertension** | | | |
| No | ref | ref | ref |
| Yes | < 0.001 | 2.21 | (1.60,3.06) |
| **Race** | | | |
| Non-Hispanic White | ref | ref | ref |
| Non-Hispanic Black | 0.21 | 0.83 | (0.63,1.08) |
| Mexican American | 0.09 | 0.46 | (0.34,0.64) |
| Other Hispanic | 0.11 | 0.6 | (0.42,0.87) |
| Other Race - Including Multi-Racial | 0.35 | 0.91 | (0.62,1.35) |

*Abbreviations*: *Ref* Reference, *SBP* Systolic Blood Pressure, *DBP* Diastolic Blood Pressure, *PIR* Poverty Income Ratio, *WC* Waist Circumference, *CR* Creatinine, *UA* Uric Acid, *ALB* Albumin, *BUN* Blood Urea Nitrogen, *TCHOL* Total Cholesterol, *HbA1c* Glycated Hemoglobin, *OR* Odds Ratio, *CI* Confidence Interval

hypertension, smoking, and TCHOL ranked at the top, followed by SBP and age, with BUN, PIR, and WC contributing relatively less. Given that XGBoost's overall performance was also excellent, we provided its feature importance ranking in Supplementary Figure S1; the top-ranking features of the two models were highly overlapping, with both identifying hypertension, smoking, and TCHOL as core predictors, indicating that key clinical risk factors have consistent and stable predictive value across different model families.

### Preliminary application of the model

To facilitate model visualization and preliminary application, we developed an online prediction platform using the R Shiny framework, deployed at https://zzaakk.shinyapps.io/make_web-1/. The platform integrates the previously established SVM model, allowing users to input eight key predictive variables: age, WC, SBP, PIR, BUN, TCHOL, smoking status, and hypertension status.

Upon entering the variables, the system generates real-time, individualized ASCVD risk predictions. To ensure user privacy and data security, the model's logic is executed locally on the client-side, and no personal data is uploaded or stored during the process. This tool offers a convenient approach for ASCVD risk assessment in individuals with diabetes and pre-diabetes, holding potential value in primary care and remote health management settings.

## Discussion

This study, based on nationally representative NHANES data, focused on individuals with diabetes or prediabetes, a population at high risk for ASCVD. Eight key predictors—age, waist circumference, poverty-income ratio, blood urea nitrogen, total cholesterol, systolic blood pressure, hypertension, and smoking status—were identified through univariate and multivariate logistic regression and incorporated into nine machine learning models for systematic comparison. The SVM model achieved the best performance in both the training and validation sets, consistently outperforming other models in terms of discrimination, calibration, and net clinical benefit. Feature importance analysis highlighted hypertension, smoking, and total cholesterol as the most influential predictors, underscoring the importance of blood pressure control, smoking cessation, and lipid management in cardiovascular prevention. Decision curve analysis further demonstrated greater net clinical benefit for the SVM model, supporting its utility in individualized ASCVD risk prediction and optimized intervention strategies.

The eight independent risk factors identified in this study are all pathophysiologically linked to the development of ASCVD and carry significant clinical implications. Elevated WC, as a marker of abdominal obesity, reflects insulin resistance, dyslipidemia, and chronic inflammation[28, 29], which collectively lead to endothelial dysfunction and plaque instability, thereby increasing ASCVD risk[30, 31]. Advancing age reduces vascular elasticity and impairs endothelial function,

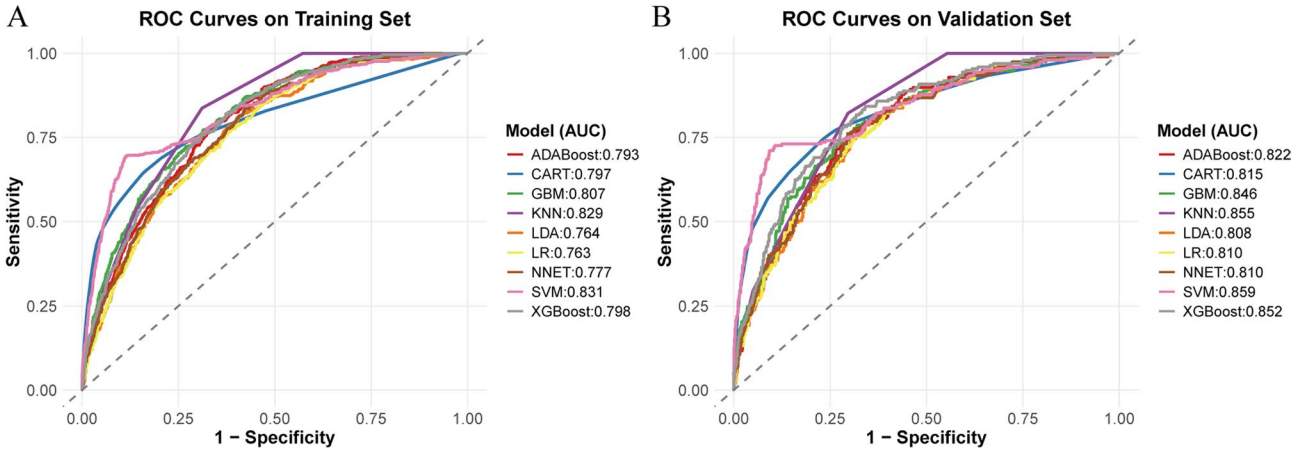



**Fig. 2** Comparison of ROC curves for nine machine learning models in the training and validation sets. **A** ROC curves in the training set, illustrating the discriminative ability of each model. **B** ROC curves in the validation set

**Table 3** Machine learning performance

| Model | Train_AUC | Test_AUC | Sensitivity | Specificity |
|---|---|---|---|---|
| LR | 0.763 | 0.810 | 0.789 | 0.653 |
| CART | 0.797 | 0.815 | 0.717 | 0.621 |
| SVM | 0.831 | 0.859 | 0.790 | 0.696 |
| KNN | 0.829 | 0.855 | 0.771 | 0.683 |
| NNET | 0.777 | 0.810 | 0.794 | 0.673 |
| GBM | 0.807 | 0.846 | 0.763 | 0.641 |
| XGBoost | 0.798 | 0.852 | 0.781 | 0.685 |
| AdaBoost | 0.793 | 0.822 | 0.772 | 0.615 |
| LDA | 0.764 | 0.808 | 0.687 | 0.623 |

*Abbreviations*: *LR* Logistic Regression, *CART* Classification and Regression Tree, *SVM* Support Vector Machine, *KNN* K-Nearest Neighbors, *NNET* Neural Network, *GBM* Gradient Boosting Machine, *XGBoost* Extreme Gradient Boosting, *AdaBoost* Adaptive Boosting, *LDA* Linear Discriminant Analysis

**Table 4** Brier scores

| Model | Training set | Validation set |
|---|---|---|
| CART | 0.096 | 0.093 |
| KNN | 0.106 | 0.099 |
| GBM | 0.109 | 0.099 |
| LR | 0.116 | 0.108 |
| SVM | 0.119 | 0.106 |
| NNET | 0.114 | 0.108 |
| AdaBoost | 0.113 | 0.107 |
| LDA | 0.117 | 0.108 |
| XGBoost | 0.111 | 0.099 |

*Abbreviations*: *LR* Logistic Regression, *CART* Classification and Regression Tree, *SVM* Support Vector Machine, *KNN* K-Nearest Neighbors, *NNET* Neural Network, *GBM* Gradient Boosting Machine, *XGBoost* Extreme Gradient Boosting, *AdaBoost* Adaptive Boosting, *LDA* Linear Discriminant Analysis

and is frequently accompanied by comorbidities such as hypertension, diabetes, and dyslipidemia, further

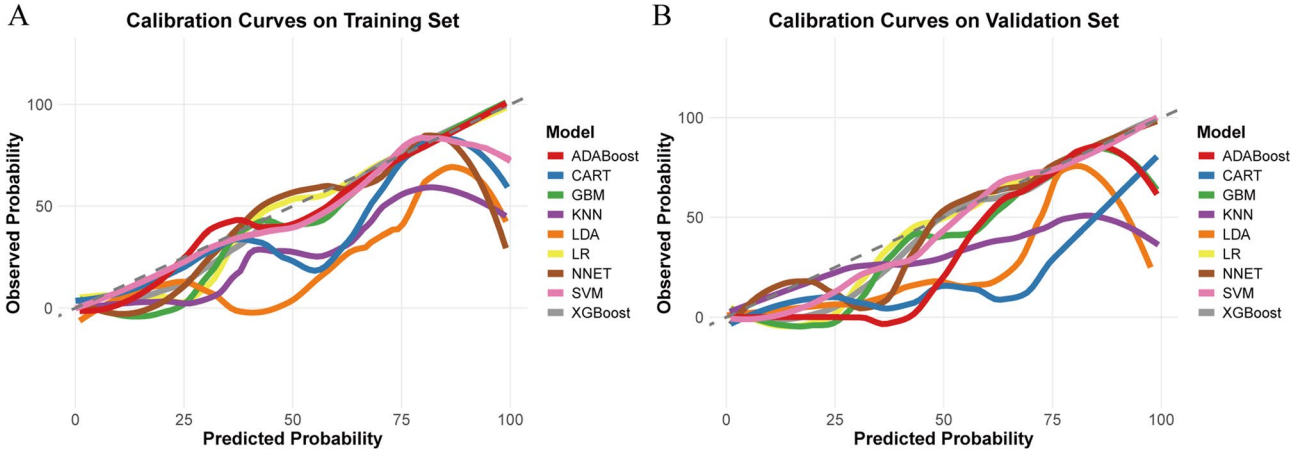



**Fig. 3** Calibration curves of nine machine learning models on the training and validation sets. **A** Calibration curves on the training set. **B** Calibration curves on the validation set. The closer the curve is to the diagonal reference line, the better the model's calibration performance Overall predictive error was measured by the Brier score (Table 4), an indicator that reflects the mean squared error between predicted probabilities and actual outcomes, where lower values indicate smaller overall prediction error. CART had the lowest Brier score (Training/Validation: 0.096/0.093), but its performance in discrimination and calibration was average, limiting its practical value. The Brier scores for XGBoost and SVM were 0.111/0.099 and 0.119/0.106, respectively, both at a good level

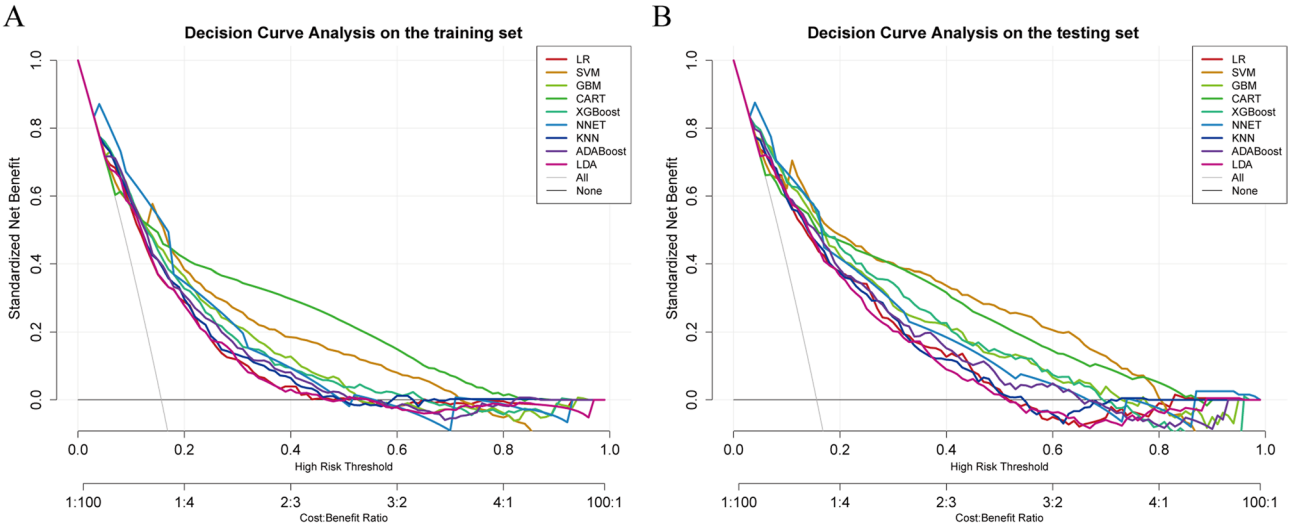



**Fig. 4** DCA of nine machine learning models on the training and validation sets. **A** Decision curves on the training set, illustrating the net benefit of each model across different risk thresholds. **B** Decision curves on the validation set. The closer a curve is to the top right, the greater the clinical net benefit of the model at the corresponding threshold

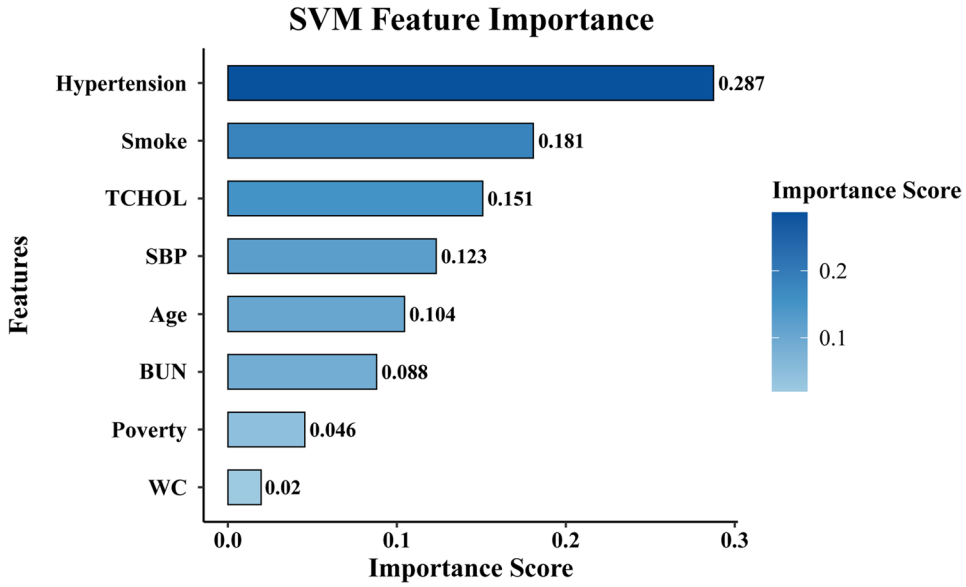



**Fig. 5** Feature importance ranking based on the SVM model. The bar chart illustrates the relative contribution of each variable to ASCVD risk prediction

elevating susceptibility[32–34]. A lower PIR indicates socioeconomic disadvantage, limiting access to healthy diets, regular exercise, and medical care, which weakens ASCVD prevention and management [35–37]. Elevated BUN suggests potential renal dysfunction, which accelerates atherosclerosis through waste accumulation and blood pressure elevation[38–40]. TCHOL is a key driver of atherosclerosis, particularly through oxidative modification of LDL-C, promoting plaque formation and acute cardiovascular events[41, 42].

Hypertension, one of the most established risk factors, directly damages the endothelium and promotes atherosclerosis[13, 43], but its long-term management is often limited by poor adherence [44, 45]Smoking significantly increases ASCVD risk by inducing vasoconstriction, elevating blood pressure, disrupting lipid metabolism, and provoking inflammatory responses [46–48].

These risk factors do not act in isolation but frequently interact and exert synergistic effects. For example, increased WC is often accompanied by hypertension and dyslipidemia [49, 50], while low PIR may influence multiple dimensions of health behaviors[51, 52].Risk assessment should therefore account for such interactions and their dynamic changes at the individual level. Notably, although these factors are well established in the general population, their detrimental impact is more pronounced

in individuals with diabetes or prediabetes. Given the underlying metabolic dysregulation and endothelial dysfunction, the same level of exposure confers higher cardiovascular risk, reflecting an"additive effect." Thus, this study not only reconfirms the role of traditional risk factors but also highlights their amplified impact in high-risk subgroups. By incorporating additional predictors such as PIR and BUN, it reveals risk differentials that general population models fail to capture, underscoring the necessity and clinical value of tailored prediction models for individuals with diabetes and prediabetes.

Unlike previous studies based on general populations (Supplementary Material S4), this study specifically focused on individuals with diabetes and prediabetes, a high-risk group for ASCVD that is often underrepresented or overlooked in existing risk prediction models. Recognizing the limitations of traditional statistical approaches in capturing complex nonlinear relationships among multiple variables, we applied machine learning methods to develop a more precise and efficient risk assessment model. Eight key predictors identified through regression analyses were incorporated into multiple machine learning algorithms, which were systematically compared using ROC curves, calibration plots, and DCA to comprehensively evaluate model performance across discrimination, calibration, and clinical net benefit. The results demonstrated that the SVM model, owing to its stronger ability to capture complex variable interactions, significantly outperformed traditional logistic regression in overall performance. Moreover, feature importance analysis highlighted hypertension, smoking status, and dyslipidemia as the most critical predictors in this population, providing essential evidence for individualized risk assessment and targeted clinical intervention.

In addition, our SVM model integrates multiple key risk factors, such as hypertension, smoking, and dyslipidemia, providing an effective tool for ASCVD risk assessment. Clinicians can evaluate a patient's ASCVD risk based on the combination and severity of these risk factors and develop appropriate intervention strategies. For patients with multiple high-risk factors (such as hypertension and smoking), more aggressive management may be required, even in the absence of clinically evident cardiovascular disease. To facilitate clinical application, we also developed an online risk prediction platform that provides real-time ASCVD risk assessments based on individual input data, helping clinicians identify high-risk populations early and optimize intervention strategies. Through this tool, our model offers practical support for personalized cardiovascular disease prevention.

Nevertheless, this study has several limitations. First, its cross-sectional design precludes causal inference. Second, the definitions of diabetes and ASCVD were primarily based on self-reported information, which may introduce recall or misclassification bias. Third, excluding participants with missing data could lead to selection bias, and residual confounding may persist despite statistical adjustments. In addition, relying solely on statistical significance for variable selection may have overlooked predictors with important clinical relevance. Finally, the absence of a non-diabetic control group limits the ability to evaluate the specificity of risk factors. Future studies should include external validation and longitudinal analyses to further confirm the robustness and clinical applicability of our findings.

## Conclusion

This study successfully developed and validated multiple machine learning models, with SVM showing the best performance in predicting ASCVD risk among individuals with diabetes or prediabetes. The model identified WC, age, PIR, BUN, TCHOL, SBP, and hypertension as key predictors, emphasizing the importance of lipid control and abdominal obesity management in clinical practice. These findings provide an effective tool for individualized ASCVD risk assessment and precision prevention in high-risk populations, supporting early screening and intervention strategies in cardiovascular care.

**Abbreviations**

| | |
|---|---|
| ASCVD | Atherosclerotic Cardiovascular Disease |
| NHANES | National Health and Nutrition Examination Survey |
| CART | Classification and Regression Tree |
| SVM | Support Vector Machine |
| GBM | Gradient Boosting Machine |
| AUC | Area Under the Receiver Operating Characteristic Curve |
| DCA | Decision Curve Analysis |
| KNN | k-Nearest Neighbors |
| TCHOL | Total Cholesterol |
| WC | Waist Circumference |
| ADA | American Diabetes Association |
| HbA1c | Glycated Hemoglobin |
| PIR | Poverty-Income Ratio |
| SBP | Systolic Blood Pressure |
| DBP | Diastolic Blood Pressure |
| BMI | Body Mass Index |
| HDL | High-Density Lipoprotein Cholesterol |
| CDC | Centers for Disease Control and Prevention |
| LR | Logistic Regression |
| AdaBoost | Adaptive Boosting |
| XGBoost | Extreme Gradient Boosting |
| LDA | Linear Discriminant Analysis |
| NNET | Neural Network |
| CIs | Confidence Intervals |
| ORs | Odds Ratios |
| TG | Triglycerides |
| CR | Creatinine |
| UA | Uric acid |
| ALB | Albumin |
| PA | Physical activity |
| AST | Aspartate aminotransferase |
| BUN | Blood urea nitrogen |

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12872-025-05247-w.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

**Acknowledgements**
N/A.

**Patient consent statement**
Study protocols for NHANES were approved by the NCHS ethnics review board (Protocol #2011–17, https://www.cdc.gov/nchs/nhanes/irba98.htm). All study participants, in the NHANES data we utilized, provided informed consent prior to their participation in the NHANES survey, as per the NHANES protocol.

**Authors' contributions**
Yang Li contributed to the study conception and design, performed data analysis and interpretation, drafted the main manuscript, and prepared the figures and tables. Bing Wang supervised the overall study design, critically reviewed and revised the manuscript for important intellectual content, and provided methodological guidance throughout the project. All authors participated in data interpretation, contributed to manuscript editing, and approved the final version of the manuscript.

**Funding**
This work was supported by the Health Commission of Henan Province through the following projects: Key Technologies and Prevention/Treatment Standards for Diabetic Foot in Elderly Patients (First Edition) (Grant No. ky2024003, awarded to Dr. Bing Wang), Specialty Construction of Peripheral Vascular Intervention Center (2023 Provincial Healthcare Service Capacity Enhancement Program) (Grant No. ky2023005, awarded to Dr. Bing Wang), and Clinical Application Study of Drug-Eluting Stents in Treating Femoral-Popliteal Artery Occlusive Disease Caused by Arteriosclerosis Obliterans (Grant No. LHGJ20210495, awarded to Dr. Yang Li).

**Data availability**
The data that support the findings of this study are openly available in National Health and Nutrition Examination at https://www.cdc.gov/nchs/nhanes/index.htm.

## Declarations

**Ethics approval and consent to participate**
The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study was conducted in accordance with the Declaration of Helsinki (as revised in 2013). All information.
from the NHANES program is available and free for public, so the agreement of the medical ethics committee board was not necessary. All the participants signed the written informed consent before participating in the study.

**Consent for publication**
All authors agree to publish.

**Competing interests**
The authors declare no competing interests.

Received: 31 March 2025 / Accepted: 1 October 2025
Published online: 04 December 2025

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